

09

Read the following passage.

Penicillin-resistant *Staphylococcus aureus* (PRSA) bacteria have evolved by natural selection. From 1941, penicillin was used extensively to treat *S. aureus* infections. The first case of PRSA was reported in 1942. By the late 1960s, more than 80% of *S. aureus* bacteria were resistant to penicillin.

Many antibiotics have a beta-lactam ring in their chemical structure, which inhibits the synthesis of bacterial cell walls. PRSA have an allele of a gene *blaZ* that codes for an enzyme called beta lactamase. This enzyme hydrolyses beta-lactam rings in antibiotics such as penicillin. 5

The evolution of PRSA resulted in the development of new antibiotics, such as methicillin in 1960. However, by 1961, methicillin-resistant *S. aureus* (MRSA) were identified. In 2022, 675 cases of MRSA were reported in England. However, in 2023, the number of cases of MRSA rose to 787. Patients in hospitals are at greater risk of MRSA infections than the general population. 10

Use the information in the passage and your own knowledge to answer the following questions.

09.1

Penicillin-resistant *Staphylococcus aureus* (PRSA) have evolved by natural selection (lines 1–2).

Name the type of natural selection.

[1 mark]

09.2

By the late 1960s, more than 80% of *S. aureus* were resistant to penicillin (lines 3–4).

Explain why.

[3 marks]



0 9 . 3

Many antibiotics have a beta-lactam ring in their chemical structure, which inhibits synthesis of bacterial cell walls (lines 5–6).

Suggest **two** reasons why this makes infections caused by PRSA difficult to treat.

[2 marks]

1 _____

2 _____

0 9 . 4

In 2022, the population of England was 5.644×10^7

In 2023, the population of England was 5.672×10^7

Use information in lines 10–12 to calculate the increase in MRSA cases per 100 000 population.

Show your working.

[2 marks]

Increase in MRSA cases per 100 000 population _____

Question 9 continues on the next page

Turn over ►



Do not write
outside the
box

0 9 . 5

Patients in hospitals are at greater risk from MRSA infections than the general population (lines 12–13).

Suggest **and** explain **two** reasons why.

[2 marks]

1 _____

2 _____

10

END OF QUESTIONS

